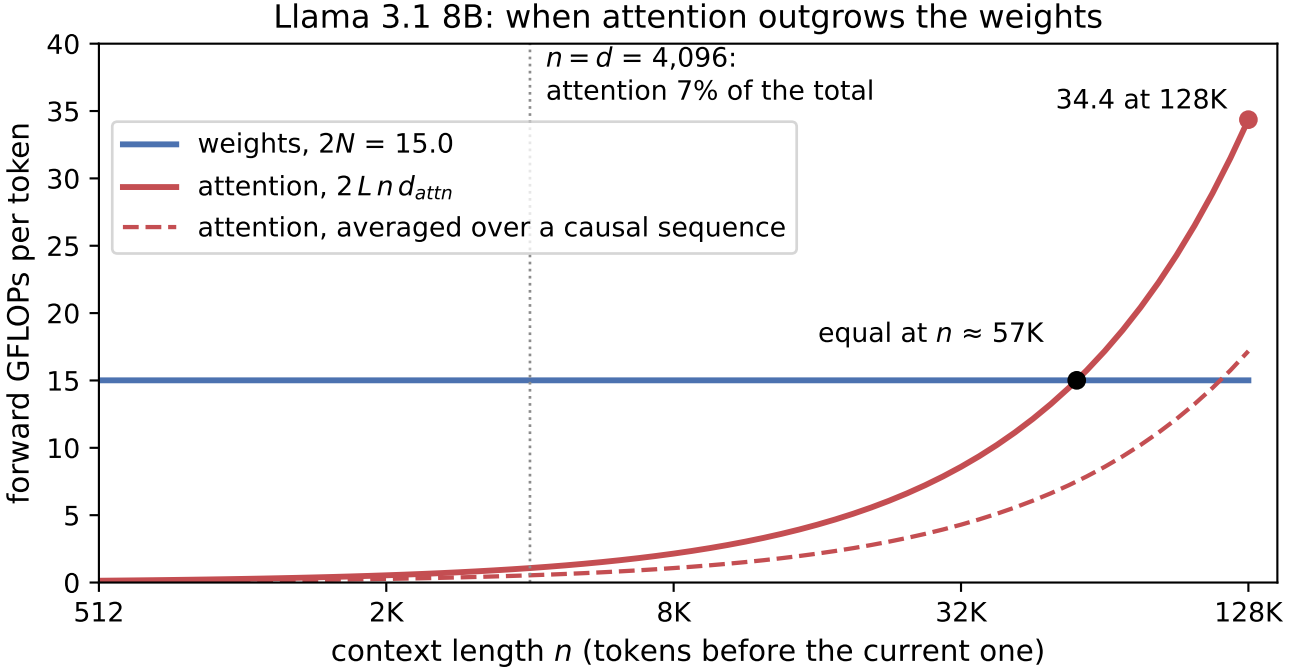
Llama 3.1 8B: when attention outgrows the weights
$n = d$ = 4,096:
attention 7% of the total
34.4 at 128K
weights, $2N$ = 15.0
attention, $2\,L\,n\,d_{attn}$
attention, averaged over a causal sequence
equal at $n \approx$ 57K
forward GFLOPs per token
0
5
10
15
20
25
30
35
40
512
2K
8K
32K
128K
context length n (tokens before the current one)